
Mercury’s Magnetosphere

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1 MERCURY'S MAGNETIC FIELD

Definition 1.1: Magnetic Regions

The regions of distinct magnetic fields are largely dictated by the *solar wind*. The solar wind is a "*stream*" of particles following the *interplanetary magnetic field* (IMF), which is largely driven by the sun.

Mercury's magnetic field is driven by currents of magma, due to the Coriolis effect, in Mercury's core that produces a magnetic field. This magnetic field is "squished" by the magnetic pressure caused by the solar wind and the IMF on the day side of the planet.

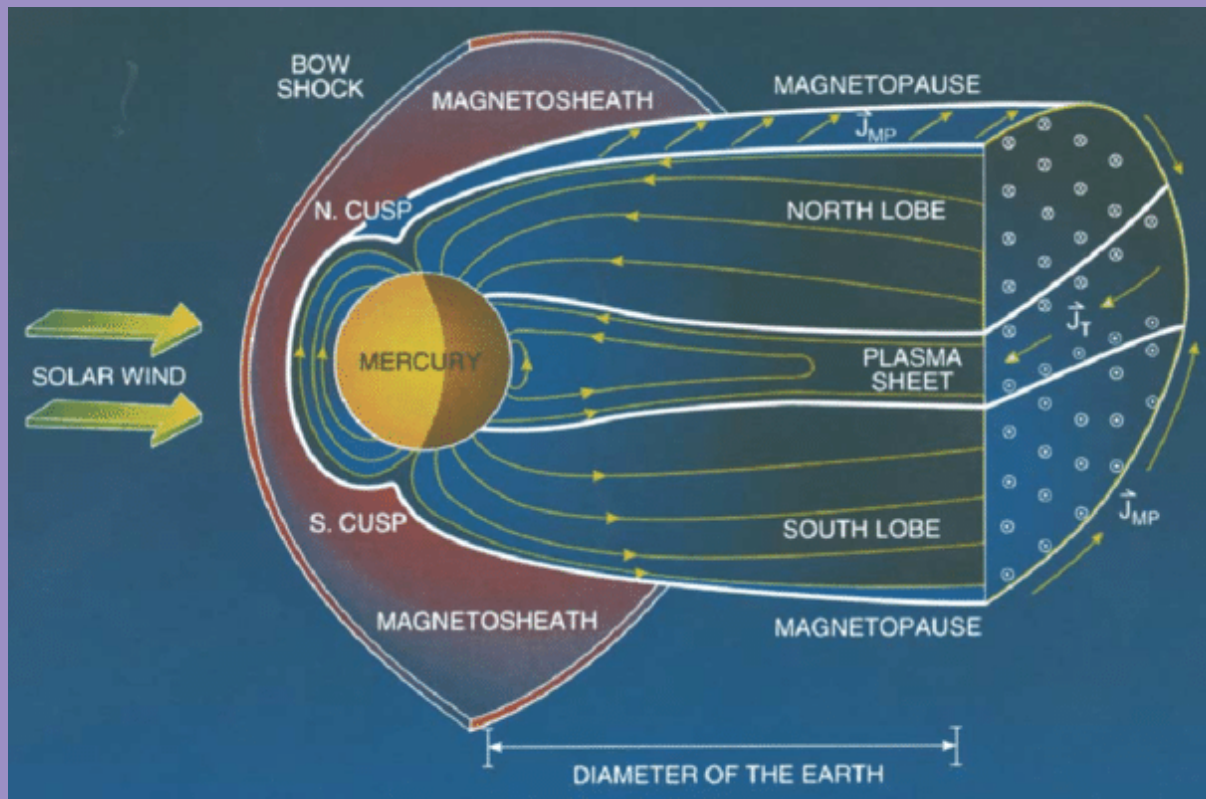


Figure 1.1.1: Diagram of Mercury's magnetic field and its interactions, showing the regions of the solar wind, magnetosheath, and magnetosphere.

Definition 1.2: Local Time

Positional value telling you Mercury's *orientation* with respect to the sun i.e 12 pm noon, is facing the sun and 12 am midnight is facing away. (**Ask what this is relative to**).

Definition 1.3: Reconnection & the Dungey Cycle

Reconnection is the process of energy transfer between the day-side of Mercury's magnetosphere, its night-side and the IMF. It refers to the process of magnetic field lines of the IMF and Mercury's magnetic field, breaking away and joining together. This leads to flux transfer events (FTE) where the magnetic field lines *reconnect* on the night-side of Mercury. This causes the Northern lights on Earth.

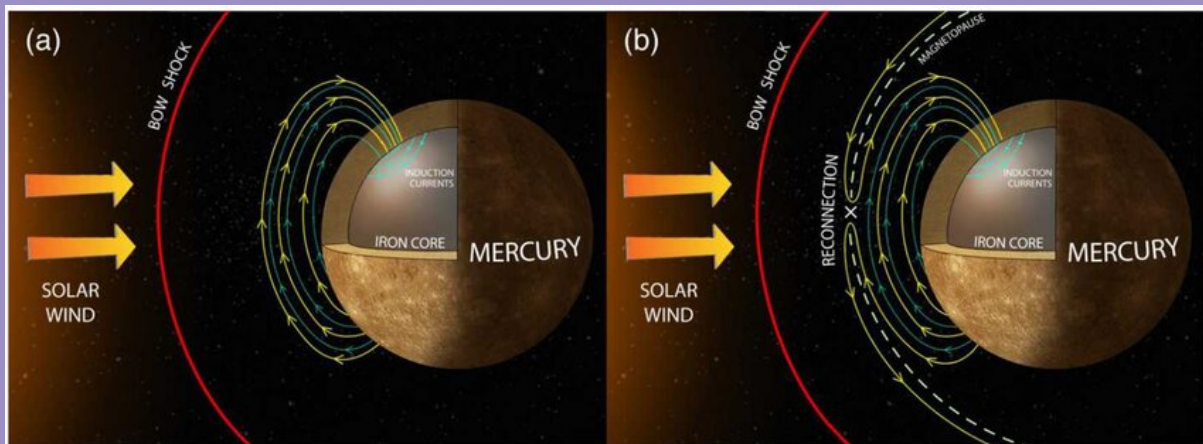


Figure 1.3.1: Reconnection happening on Mercury. The point where reconnection begins is called the X-line.

The *Dungey cycle* is the timescale in which reconnection happens for a given system, on the order of minutes to hours.

Definition 1.4: Parker Spiral

The "bunging up" of the sun's magnetic field lines due to its spin.

Definition 1.5: Satellite Motion Terminology

Solar Wind	→	Magnetosheath	Bow shock in (BS_IN)
Magnetosheath	→	Solar Wind	Bow shock out (BS_OUT)
Magnetosheath	→	Magnetosphere	Magnetopause in (MP_IN)
Magnetosphere	→	Magnetosheath	Magnetopause out (MP_OUT)

Always relative to magnetic field of Mercury.